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From quantification to methodolatry

Twentieth-century psychology's infatuation with numbers would not have blossomed into an all-consuming love affair had it not been for the usefulness of statistical constructions based on group data. Statistical constructions represent work done on some raw material that already has a numerical form. However, that form involves more than the counting of heads. The individuals who are counted must also be endowed with countable attributes. Where the attributes are physical, like height or weight, their transformation into numerical form depends on social practices developed long ago, but where one has to deal with psychological attributes the corresponding measuring practices represent a relatively recent development. This development was initially independent of the special use of statistical information that has occupied our attention so far. It is therefore necessary to step backward in time to consider the nature of the practices that made it possible to impose a numerical form on psychological attributes.

What is the difference between one and two?

Because it is rare for human experience and activity to take on a quasi-numerical form spontaneously, psychologists who insist on working with such forms must impose them on their subject matter. One way of doing this is to apply the numerical net post hoc, after the psychological event has taken place and has produced some record. The number of word associations with a particular thematic content can be counted, for instance, either for a particular individual or across individuals. This kind of quantification was attempted by Galton and practiced on a large scale by G. Stanley Hall.¹ Here the production of numerical data depended on the researcher establishing certain thematic categories and then performing a

number of equivalence judgments that resulted in unambiguous decisions about the inclusion or exclusion of each judged item for the categories employed. The point is that the numerizing operations were clearly performed by the researcher and not by the human source of the data. The latter's experience and activity was not directly affected by these operations, except in the special case where the researcher was his own data source.

This approach, however, was of limited significance for the subsequent development of quantifying practices in psychology. Much more important was a second approach that attempted to impose a numerical structure on the data of psychology right at their source. The various techniques enlisted in the service of this aim can be regarded as constituting a hierarchy of increasingly complete control of individual behavior. First, a numerically graded stimulus series can be prepared – for instance, a series of weights of increasing heaviness – and the subject's responses to each member, or each pair of members, of the series can be recorded. This was done very early in G. T. Fechner's psychophysical methods,² which formed a mainstay of the work of the first psychological laboratories.

However, it should be noted that for a quantitative psychology something more is needed than a numerically graded stimulus series. It is also necessary to limit the subject's responses to these stimuli. A qualitative description of sensations is not what is required here. The experimental subject must limit himself to simple judgments, and his answers must be cast into an unambiguous, digitalized form of the yes–no type. Only under these conditions is it possible to obtain a quantitative function that links the stimulus series to the response series in the kind of relationship that Fechner called Weber's Law. For instance, the subject must agree to limit his responses to alternatives like “heavier versus lighter” or “equal versus unequal” if the goal of this sort of investigation is to be achieved. Because Fechner experimented on himself, the goals of the investigator and the duties of the experimental subject were brought together in a kind of personal union, but in a social functional sense they are distinct. There is an indispensable element of social control wherever the form of subject responses is strictly subordinated to the goals of the investigation, although the social aspect may well be hidden by the voluntary nature of the agreement and by the impersonal nature of the requirement.

Sometimes the practice of controlling the form of the subject's response in the interests of quantification did lead to debates among experimentalists in which the social basis of the practice emerges clearly enough. Numerization required that only responses of the dichotomous yes–no type be given legitimacy. But what was one to do about human individuals' spontaneous and persistent tendencies to break out of this straightjacket, if only by giving responses in the “doubtful” category? There were two alternatives. One could prevent the subject from taking such liberties by

insisting that all answers be either yes or no (the method of right and wrong cases), or one could be more permissive and allow doubtful responses, but afterward manipulate the data so that the effect of such responses was washed out.³ The simplest way of doing this was Fechner's method of dividing the "doubtful" items among the legitimate (i.e., yes or no) categories. Whatever technical sophistication was added in the latter case, there had to be a basic decision on how far the investigator ought to push his authority in the experimental situation. Such an issue could only arise within an existing social framework. The production of quantitative data involved social practices of investigation at a very fundamental level.

More generally, social numerizing practices of the early experimentalists linked theory and data in a tight methodological circle. For Fechner, and those who followed him, the confinement imposed on the subjects' responses seemed appropriate and justified because "sensations," which was what they considered themselves to be studying, were assumed to constitute psychological atoms or elements that could be incrementally added if they were of the same kind.⁴ The instructions and response categories imposed on individuals acting as the data source were not thought of as arbitrary by investigators who were inclined to believe in an elementaristic theoretical model in any case. But when they gave expression to this belief in the limitations they imposed on the data source, they inevitably caused it to produce data that were consonant with their own assumptions.

Data produced under such rules were not permitted to contradict the assumptions on which the rules were based. Instead, it became customary to introduce pseudoexplanations when the social practice of forced numerization produced uncomfortable results. For instance, in investigations of the tactile two-point discrimination threshold so-called paradoxical judgments were common. The subject's skin would be stimulated by either a single point or by two points some distance apart in an attempt to measure tactile sensitivity. Without the help of visual cues, the subject then had to say whether he or she *felt* one or two points. Now, while it was expected that when the two stimulated points were too close together only one point of stimulation would be felt, it was not expected that a single stimulated point would be reported as feeling like two. Yet that is exactly what happened in a significant number of cases.⁵ Moreover, the proportion of such "errors" tended to decrease (often quite massively) as the experiments were repeated. This result meant that the numerical value of the sensitivity threshold could vary widely, depending on when in the experimental series it was calculated.

Such results generally led to a search for technical improvements and auxiliary hypotheses rather than to any questioning of the fundamental practices on which these attempts at numerization rested. Reductions in "paradoxical" responses were said to be due to the effects of "practice."

This was hardly an explanation, for it left dark what exactly the word "practice" referred to.

One psychologist who was prepared to probe further, and who achieved an unusual degree of insight into the basis of psychophysical numerizing practices, was Alfred Binet. The first suggestion of what might be going on was provided by his fifteen-year-old daughter, Madeleine, whom he had put through a number of these experiments, only to find the usual "practice" effects. "I knew better this time what the sensations meant," she told him after a later session. "When a sensation was a little 'big,' I thought there must be two points, because it was too thick for one."⁶ What Madeleine had picked up was the rather elastic meaning of the category "two" that had to be employed in this situation. Fortunately, her father had no particular stake in the psychophysical methods pioneered by his German rivals and decided to pursue the matter. He extended the investigation to adult subjects, permitting them to report freely on their experiences. Under these conditions Fechner's neat binary scheme of *either one or two* disintegrated. Subjects reported sensations between oneness and twoness – "broad," "thick," and "dumbbell-shaped" sensations that could be categorized as "one" or "two," depending on the subject's interpretation of the schema that the experimenter wished to be applied.⁷ Binet concluded that the threshold of two-point discrimination could not be scientifically determined.

But such radical conclusions were the exception rather than the rule. Subsequent investigations did lead to a recognition of the qualitative complexity of the tactile sensations on which the concept of a measurable two-point discrimination threshold had to be imposed. But the constant feature of the various introspective descriptions of the underlying tactile experiences was their *arithmomorphic* form.⁸ These experiences always tended to be described in the form of distinct patterns that formed a series between oneness and twoness, such as point, circle, line, dumb-bell, two.⁹ Of course, this simply meant that intermediate categories like "point-or-circle" now had to be added, but if one wanted to take the trouble one could go on to determine thresholds between all the adjacent categories in this series. By insisting on distinct categories, and by arranging them in a series, one had already imposed a protoarithmetic form on one's experience, a proceeding that reduced the actual operations of measurement to a purely technical matter. One problem that emerged with the attempt to impose more complex series on the original experience was the lack of agreement among investigators on the number (and quality) of the categories that constituted the series. If one investigator finds five categories between oneness and twoness, and another eight different categories, the arbitrariness of the whole procedure does become rather obvious. The simple two-point threshold did at least have the crucial advantage that a social

consensus could be achieved on the basis of the features of the physical instrument of stimulation that was common to all investigators and that indeed had two points.

However, before this physical twoness could be translated into the twoness of subjective sensation that was the basis of the psychological measurement, a little social construction had to be performed. An agreed-upon definition of "two" had to be arrived at by the participants in the investigation. Were tactile sensations that were verbally described as "point-or-circle" or "elongated oval" to be reported as "one" or as "two"? For Madeleine Binet it might simply be a matter of obliging her father, but with adult subjects there was more room for misunderstanding, especially when the usual cultural pre-understanding of the requirements of the situation could not be taken for granted. Thus, when the famous anthropological expedition to the Torres Straits added a new dimension to experimental psychological research by attempting to make psychophysical measurements on the local inhabitants, it was found that the average two-point discrimination threshold of the natives was less than half that of Englishmen. The probable reason was guessed by a canny experimenter like Titchener: The English subjects were acting on the basis of their understanding of the norm of accuracy and were refusing to call "two" any sensation that did not indubitably involve two separate points with a distance between them. But the South Sea Islanders, wanting to impress the powerful strangers by their ability to detect the presence of two points, would respond with "two" whenever their actual sensations left any room for doubt.¹⁰

In the words of E. G. Boring, "the meaning of the judgment *two* is indeterminate unless the criterion has been established."¹¹ The practice of psychological measurement was clearly grounded in the imposition of agreed-upon definitions on the raw material of individual experience. What was true of the category two in the case of the two-point discrimination threshold was surely true of the other basic judgments, like equal and unequal, on which psychophysical measurement depended.

But perhaps an objectivistic basis for psychological measurement could still be asserted if the measured sensations naturally formed a linear series, so that the socially constructed judgment categories only affected the place in the series where divisions were introduced, but did not affect the serial nature of the experiences themselves. This became the standard defense of orthodox experimentalists, but it hardly convinced their critics. True, psychologists who expected subjective sensations to be measurable were able to offer serially arranged descriptions of the sensations that fell between oneness and twoness, for example. But the skeptics countered with their own introspective examples of the qualitative distinctness of sensations that were supposed to form an incremental series.¹² This became

known as the quantity objection to psychophysics, and the discussions that it led to were remarkable for their inconclusiveness.¹³

Both sides to this debate shared a common definition of the problem of psychological measurement. It was seen as *mental* measurement, that is to say, the measurement of entirely private events. If mental measurement in this sense was impossible, then the only conceivable alternative was that what masqueraded under that name was in fact physical measurement. This was a manifestation of the notorious "stimulus error," to which psychologists were said to be prone. But the experimentalists were able to point to their success in establishing relationships between physical and psychological magnitudes, as in Weber's Law. If the sensation side of Fechner's equation was not based on mental but on physical measurement, then Weber's Law would seem to be a law of physics. Yet the basis for the claim that what was being measured were pristine private events was equally dubious. Both sides underestimated the socially constructive, stipulative, aspect of so-called mental measurement. Investigators were indeed able to construct numerizable series of judgments that were different from the operations of physical measurement, although there need be no isomorphic series of private experiences that stood in a one-to-one relationship to the public judgments.¹⁴ Private experience, with its vast potential for interpretation, set few limits to the social constructions that might be imposed on it. The psychological measurements that were the products of this construction certainly had a reality of their own, but it was not a reality that could be neatly classified in terms of the binary categories of the traditional duality of physical body and private mind.

Serializing behavior

Had the meaning of psychological measurement never extended beyond the measurement of private mental events its future would have been extremely limited. Unless one believed, like Fechner, that such measurements could make a contribution to the metaphysical mind-body problem, they were of little theoretical interest. Even the repeated ritual of demonstrating the measurability of mental events must surely have exhausted its appeal after a time. Sensory threshold measurement was of some psychophysiological interest and eventually found practical applications in human engineering. But these successes did not really depend on an answer to the question of whether private mental events were really being measured. What was important in these contexts was the possibility of giving quantitative expression to some *capacity* of the individual organism. Relevant were questions of *performance* against some externally defined criterion, not questions of the quantitative structure of inner sensory experience. Psychological measurement became a generally useful tool only

when it was interpreted as constituting a measurement of individual capacity rather than of individual experience. Apart from the jettisoning of philosophical embarrassments, this step enormously expanded the potential scope of psychological measurement, because the number of measurable capacities was limited only by the ingenuity of psychologists in devising tasks with quantified performance criteria.

The problem of psychological measurement therefore shifted from one of imposing a quantitative structure on subjective experience to one of imposing quantitative structure on human action. First of all, if measures of individual action were to function as measures of individual capacity, these actions had to be structured as *performances*. That is to say, they had to result in some product that could be graded against some imposed criterion. Then, in order to give the measurement the appearance of objectivity, an arithmomorphic form had to be imposed on the individual subject's actions. In other words, performances had to be segmented into distinct but equivalent units that could be counted. Under these conditions the numerical value assigned to the performance as a whole seemed to reflect some degree of capacity within the subject rather than the subjective judgment of the investigator.

All the fundamental features of the measurement of psychological capacity were first manifested in Hermann Ebbinghaus's classical work on memory. Using himself as subject, Ebbinghaus endeavored to measure "memory" by constructing lists of nonsense syllables that he presented to himself repeatedly.¹⁵ Whenever he desired to do so, he could attempt to reproduce the list and express the result in terms of the fraction of the original list that he had successfully reproduced. In this way his memorizing performance could be expressed in a number, and he could make quantitative comparisons of his performance under different conditions.

Ebbinghaus was not the only person who was interested in the experimental investigation of human memory around 1880. Wundt, for instance, had some interest in this question and had one of his students, the American K. H. Wolfe, work on the topic. Ebbinghaus and Wundt had one thing in common in that they both reduced the problem of memory to a problem of psychological reproduction, but this common ground makes the difference in their approach all the more striking.¹⁶ In Wundt's laboratory the experimental subject was presented with a standard tone and subsequently with comparison tones at varying intervals of time after the standard tone. In each case the subject had to judge whether the pitch of the standard and the comparison tone was the same or different. The proportion of correct judgments could then be plotted as a function of the time that had elapsed between the presentation of the standard and the comparison tone.¹⁷ What was supposedly being measured here was the ability of the experimental subject to reproduce accurately the sensory memory of the standard tone so that he could compare it with the comparison tone, an

ability that seemed to decline with the time that had elapsed since the presentation of the standard tone. For Wundt, psychological reproduction meant the reproduction of a subjective experience,¹⁸ and any attempt at quantification implied a truly mental measurement as it had for Fechner.

For Ebbinghaus, however, psychological reproduction in his famous memory experiments had a rather different meaning. Here it was not a question of whether a particular subjective sensation could be accurately reproduced but of whether a certain objective result could be achieved, irrespective of any private experience. In both cases psychological measurement depended on a matching operation, but in Wundt's case there was an attempt at direct matching of two subjective experiences, whereas for Ebbinghaus it was some public product of the individual's memorizing activity that was matched against the criterion. For Wundt, the objective stimulus was simply the occasion, and the external activity of the subject was simply the expression of the subjective process, which was the real focus of the psychologist's interest. For Ebbinghaus, the matching of the products of the subject's activity with objective requirements was itself the primary focus of experimental interest. The question was how many members of a given series of nonsense syllables could be successfully reproduced or recognized under different conditions. Ebbinghaus's methodology established the category of achieved performance (*Leistung*) as a fundamental organizing principle of experimental psychological research.¹⁹

This transition entailed certain changes in the nature of the arithmomorphic net that the investigator had to cast over his subject matter in order to come up with a product that could still be called a psychological measurement. It was no longer a question of measuring private experience but of measuring some capacity attributed to an individual, that could be held responsible for a certain level of achieved public performance. That required the juxtaposition of two countable series: the "stimulus" series that defined the task, and a series of individual actions that represented the individual's attempts at measuring up to the task. In Ebbinghaus's case the first of these was represented by his well-known series of nonsense syllables, and the second by a series of repeated attempts at memorizing, or "trials." Let us consider these in turn.

If we merely think of the nonsense syllable as constituting the crucial feature of Ebbinghaus's method, we will have missed its crucial structural feature. He himself was very clear about the fact that his instrument was not the nonsense syllable but the *series* of nonsense syllables. All his experimental questions were addressed to such series, not to individual syllables. (It is worth noting here that the specific experimental tradition to which his work led has been characterized by the serial nature of the stimulus material to which it is bound and not necessarily by the use of the nonsense syllable as such.)²⁰ All of Ebbinghaus's data were based not on a mere collection of syllables but on a specific arrangement of syllables.

The structure of this arrangement is that of a linear sequence of equivalent units with a determinate length that is countable in terms of the number of units. In fact, he tried using a series of numerical digits for his experimental purposes and only rejected this stimulus material because of some practical problems.²¹ At any rate, Ebbinghaus made sure that he had task materials with a quasi-numerical structure, so that their reproduction could lead to a countable number of matches of the constituent elements. With such a structure the counting of matches of individual elements could be considered to give a measure of the adequacy with which memory had reproduced the whole.

But that was only one side of Ebbinghaus's method. The other side involved a structuring of the experimental subject's activity so that the quasi-numerical series, which was characteristic of the stimulus material, recurred on the subjective side of the investigated relationship. The process of the experimental subject's familiarization with the stimulus material was deliberately broken up into a number of equivalent segments, called "trials," which followed one another in a linear chronological sequence. In more traditional investigations experimental subjects might also have to engage the stimulus situation repeatedly, but these repetitions did not constitute a cumulative series. They were not "trials" but repetitions of the experimental observation for the purpose of increasing reliability and decreasing observational error. This aspect also recurs elsewhere in Ebbinghaus's methodology, but of course the point of repeated attempts at memorizing a particular list lies not in mere repetition but in *cumulative* repetition. Ebbinghaus's method is therefore based on the confrontation of two arithmetically structured series, the stimulus series of nonsense syllables and the series of trials that represent the activity of memorizing. This makes it possible to establish functional relationships between what are now two sets of objectively defined variables.

The metaphor of work and the religion of numbers

What induced Ebbinghaus to adopt this approach? He was certainly no behaviorist. It appears that he had certain preconceptions about memory, amounting to a prototheory that suggested to him what sorts of data would be relevant within his conceptual framework. He seems to have started out with a basic conception of memory as a kind of *work*.²² Instead of thinking of memory as a faculty or being concerned with the reproductive aspect of inner experience, he concentrated on the activity of memorizing, which he saw as an energetic process. The work of memorizing led to an increase in the "strength" of the memory image and made it more easily reproducible. This "strength" of internal representations had nothing to do with their meaning or their quality – it was a purely quantitative, energetic property, and the relation between it and the amount of work

invested was conceived by analogy with the corresponding model in the physics of the time.²³ Working within such a framework Ebbinghaus set about constructing a measure of the amount of work invested in the activity of memorizing and a measure of the “strength,” or amount of energy accumulated in the memory trace. The former was provided by counting the number of trials, the latter by the proportion of correctly reproduced items in the series of nonsense syllables.

His procedure would have represented no more than a historical curiosity had the metaphor of psychological activity as a kind of work not had such widespread appeal. That appeal was strongest for those psychologists who were interested in making themselves useful in the context of educational institutions. In Germany the tasks facing such psychologists were commonly regarded as calling for an approach to psychological investigation that was fundamentally different from that of the usual laboratory psychology of the time. However, this difference was not seen in terms of a need for statistical information on large numbers of subjects, but in terms of a difference in the focus of investigations with individual subjects. Whereas the laboratory psychology that traced its origins to the laboratories at Leipzig and Göttingen was concerned with the analysis of elementary features of the inner experiences of adults, the new school psychology would have to work with children and would have to be much less restrictive in its focus if it was to produce anything of practical value in the classroom. Yet, school psychology was by no means equivalent to child psychology, which was concerned with general questions of psychological development. The practical tasks of school psychology seemed to require a much more specific focus. That focus was defined by Ernst Meumann, its most prominent German representative, as the child at work.²⁴ With such an object of study the investigative practices of the new subdiscipline would clearly have to rely very heavily on the pioneering work of Ebbinghaus.

Meumann, who had been Wundt's assistant for six years (1891–1897), was sufficiently steeped in the perspectives of the traditional psychology of consciousness to know that within this framework work could not be regarded as a psychological category. But he thought that when psychological processes were put in the service of “intended or prescribed success” they manifested themselves as psychological work and had to be studied as such. Under these circumstances psychological processes became a means to an end and ought to be studied from the point of view of “psychological economy” – that is, how the desired success might be achieved with the least expenditure of effort.²⁵ Put in these terms, the investigations of the psychologist were seen to be clearly relevant to the day-to-day interests of educators who wanted to increase the efficiency with which children at school succeeded at their prescribed tasks.²⁶ More specifically, these investigative interests led to the development of a new field of psychological research concerned with the so-called economy of learning, the

search for “the best means of saving time and energy” in the accomplishment of learning tasks of the type encountered in schools.²⁷

The methods most suited to the pursuit of these research goals were essentially those pioneered by Ebbinghaus. They were based on the use of two measures, a measure of success at achieving some defined task, and a measure of the work expended in achieving this success. The former was most often supplied by the use of serially ordered stimulus material and the latter invariably involved counting the number of repetitions or “trials” needed to achieve a certain level of objectively defined success.²⁸ To use such measures one had of course to impose the same kind of structure on the activity of the experimental subjects as had Ebbinghaus. But as this structure had a great deal in common with the structuring of learning activities in the schools of the time, no one was likely to regard it as inappropriate to the limited goals of this type of research.

Limited goals were utterly foreign to Meumann’s American counterpart, Edward Lee Thorndike. Both men led the way in the establishment of a quantitative psychology of school learning, and on that level the investigative practices they advocated were quite similar. What distinguished them was the enormous disparity in their influence and the striking divergence in their knowledge claims.²⁹ In the history of twentieth-century German psychology and education, Meumann remains a somewhat marginal figure. He died relatively young, but much more significant, he lacked both an institutional powerbase comparable to Thorndike’s at Columbia’s Teachers College and a group of socially effective allies like Thorndike’s school superintendents. Meumann’s audience consisted mostly of classroom teachers with an interest in rationalizing their work but with little power.³⁰ Academic psychologists on the one hand and administrators on the other saw little or no reason to take much notice of his methods. Thorndike, however, became the leading representative of an army of educational efficiency experts³¹ and played a central role in the formation of a whole new branch of psychology that was to supply the rest of the discipline with many of its basic concepts – the psychology of learning. But for that to happen his work had to be seen, by himself and by many of his academic contemporaries and successors, as having a significance far beyond the walls of the classroom.

For Thorndike, and for the substantial number of American psychologists who shared his outlook, “learning” was not just an activity characteristic of school children but a fundamental biological category applicable to everything from amoebas to homo sapiens.³² So the potential scope for a quantitative investigation of learning was not limited to the special tasks of a certain kind of educational psychology. The methodology that had served so well in the narrower context was to be extended to the large domain of the behavior of organisms in general. This extension was entirely in line with Thorndike’s metaphysics of quantification. “Whatever exists

at all exists in some amount," he proclaimed in his most famous statement, and from investigations that span half a century it is apparent that he equated science with effective measurement of these amounts.³³ Moreover, science as measurement was something worth pursuing with a truly apostolic zeal.³⁴

These beliefs were inseparable from the ambitious authority claims of those who represented quantitative science and spoke in its name. These were the experts in social management and control among whom a new breed of psychologist was to occupy a specially important place.³⁵ Quantitative data by themselves were of course just marks on paper, but they could be transformed into a significant source of social power for those who controlled their production and interpreted their meaning to the non-expert public. Quantitative psychological knowledge was a species of esoteric knowledge that was held to have profound social implications. The keepers of that knowledge were to constitute a new kind of priesthood, which was to replace the traditional philosopher or theologian.³⁶ As long as the new knowledge was used to buttress the foundations of the prevailing social order – which was invariably the case for Thorndike and those who shared his views – the experts who produced such knowledge were assured of lavish support from the centers of social power.³⁷ This was the social context in which quantification was gradually transformed from a set of means for the pursuit of specific goals into an end in itself. But on another level quantification did remain a means, a means for the achievement of social status, not only for the discipline as a whole, but for its subareas and for individuals within those subareas. The principles that were successfully invoked for the public legitimation of the discipline always tended to be reflected in the norms that governed the internal status hierarchy within the discipline.

The methodological imperative

Despite some practical difficulties and repeated doubts voiced in various quarters, the commitment of American psychology to quantitative methods was relentless. This was a flood fed from powerful sources. Some of these sources had their origin in the external relations of the discipline, and these will by now be clear enough. Quantification seemed to mark psychology as one of the exact sciences and to distinguish it sharply from such questionable pursuits as philosophy and spiritualism, with which it had been popularly associated. Moreover, as we saw in chapter 7, the first large-scale success of modern psychology as a practically useful discipline had been based on its ability to supply essentially statistical information, and this necessarily placed a premium on quantitative data. The larger legitimation claims of the discipline depended on its ability to make a social contribution that would be regarded as more valuable (i.e., more useful)

than that of its potential competitors. Reliance on precise, quantitative data would provide an impressive foundation for such claims.

These powerful forces, generated by the social context within which the discipline had to operate, should not lead us to overlook the importance of the role that quantification came to play in the internal life of the discipline. The most pervasive effects of the social context were to be seen in the elementary fact that psychology changed from a more or less amateurish interest of independent investigators to a collective enterprise organized along disciplinary lines. Thus, it had to develop a set of institutionalized rules that would regulate and coordinate the activity of its individual members so that they contributed, or at least did not visibly harm, the disciplinary project. Rules relating to quantification assumed increasing importance as the discipline grew in size and complexity and in the ambitiousness of its investigative goals.

From its earliest beginnings modern psychology had had to face the paradox that while its *subject matter* was defined as a private object, namely, the individual consciousness, the procedures by which it proposed to study this object presupposed a collective organization, namely, a community of communicating scientists. Their work was to be interdependent and cumulative, as it was in the established natural sciences. Each person's work was to build on that of others, allowing the gradual construction of an edifice of knowledge, brick by brick. That, however, would only be possible if there existed some common recognition of the rules under which this construction was to proceed and some consensus about what constituted a genuine contribution to the edifice of knowledge and what did not. Psychology as a whole had found it peculiarly difficult to achieve agreement on these matters and hence to generate a product that represented *the* truth about its subject matter by common consent. The imposition of a quantitative structure on its knowledge base had seemed to many to offer a resolution of these difficulties. Certainly, quantified items of knowledge seemed to be unambiguous and capable of commanding universal recognition.

Those advantages of quantitative data were however entirely dependent on the possibility of achieving consensus with regard to the rules that governed their production on the one hand, and their interpretation on the other. One might say that the preference for quantitative data merely shifted the problem of achieving consensus onto different, though perhaps more promising, ground. If the ground was indeed more promising, this would presumably be due to the superior prospects, in a numerizing civilization, of gaining adherence to conventions governing the production and interpretation of numerical information.

In the first section of this chapter we explored the fundamental dependence of the process of generating quantitative psychological information on the operation of certain social conventions. Although, as we saw, critical

voices were by no means absent when these conventions began to be widely applied, they were quite ineffective in undermining the common faith that what was achieved by adherence to these conventions was more than a merely conventional truth. By the second decade of the twentieth century, the ground rules for the *production* of numerical psychological data were well enough established to be utterly taken for granted. Any debate regarding fundamentals now began to shift to the second major issue, the *interpretation* of quantitative information. It may have been relatively easy, given a particular cultural context, to secure adherence to the conventions that made possible the production of quantitative psychological data, but that still left open the tricky question of what such data meant – in particular, what general inferences one could legitimately draw from such data.

In view of the role that statistical procedures had played in the history of quantitative psychology, it was perhaps inevitable that this question should present itself as a question of *statistical* inference. The Galtonian turn of psychological investigation, discussed in chapters 5 and 7, necessarily multiplied data that were essentially statistical, based on the aggregation of information from many individuals. But even before this development, it had only been possible to detect regularities in the numerized responses of individuals by treating them statistically. This was true of Fechnerian psychophysics as well as of the research tradition begun by Ebbinghaus. Quantitative data in psychology generally meant data that had been statistically arranged. To ask what consensual agreement might be achieved on the basis of such data was therefore easily seen as being equivalent to asking what statistical inferences had a claim to universal assent. This way of putting the question had the advantage of converting a fundamental question about the scientific project of psychology into an essentially technical question that could be solved by the adoption of a few conventions governing the use of certain technical devices. The work of producing quantitative data could then proceed as though there were no unanswered questions about the psychological reality that these data represented.

This pragmatic attitude was clearly the appropriate one in the context of an “applied” psychology that sought answers to practical problems that demanded administrative decisions. Indeed, this is the context in which the attitude can be documented at a very early stage. As outlined in chapter 7, a new era in psychological experimentation had been ushered in by the English school inspector, W. H. Winch, who attempted to improve the efficiency of instruction by comparing the relative improvement of performance in groups of school children subjected to different experimental treatments. There was however one major deficiency in Winch’s earliest studies. They were statistically naive and attempted to base general conclusions on a mere comparison of averages. The defect was quickly spotted

at University College, London, the fountainhead of the new biometric techniques of statistical analysis. Under the auspices of Charles Spearman, an extensive criticism of Winch's work was soon published in which it was pointed out that averages had to be compared with their "probable errors" before any conclusions could be reached about the "significance" of any difference between them.³⁸ Statistical techniques for performing this operation were available, so that the question of deciding on the generality of the findings could be reduced to the question of deciding on how many times a difference between means had to exceed the probable error of the difference in order to constitute "true significance." That could be settled by agreeing on a conventional value.

Exemplary pragmatist that he was, Winch immediately saw the point of the criticism and began to adopt the recommended statistical devices in his later studies.³⁹ Gradually, the estimation of statistical significance became standard practice in the educational psychological literature, replacing the more informal methods of group comparison that had been in vogue previously.⁴⁰ Once the practice had become general there was an obvious need to find a shorthand term to refer to the comparison of a statistical difference to an estimate of its variability. That need was filled by the term "critical ratio," which had originally been introduced in the context of comparing the effects of different types of fiscal control of school systems.⁴¹ From the area of educational administration the use of the term quickly spread to the closely related area of educational psychology and thence to general psychology. By the mid-1930s the use of "critical ratios" was an essential feature of much psychological research and merited extended treatment in textbooks of psychological statistics.⁴² A search of four prominent journals (*Journal of Experimental Psychology*, *American Journal of Psychology*, *Journal of Abnormal and Social Psychology*, *Journal of Applied Psychology*) indicated that by the period 1934-1936 about one-third of all empirical papers based on group data also reported critical ratios.⁴³ Most of the rest used correlational techniques.

For practitioners like Winch, such estimates of the significance of differences had made immediate sense because it was believed that they indicated the probability of obtaining a difference as large as the obtained one if the experiment were repeated a large number of times. That was a good basis for making a practical decision about whether it would be worthwhile changing instructional programs on the basis of a limited set of experimental results. However, the early investigators in educational psychology were never content with a simple statement of practical expediency but always claimed that their statistical findings reflected on the nature of fundamental human functions or capacities, like memory or intelligence. Such claims necessarily implied rather more complex issues, for, as the practice of statistical significance testing became widespread in psy-

chological research, the issue that would ultimately have to be faced was that of the relevance of statistical data for psychological theory.

E. G. Boring, the future historian of experimental psychology, appears to have been the first American psychologist to become seriously concerned about this question. It is likely that his concern was part of a more general interest in defending the role of purely scientific psychologists against the spreading influence of applied psychologists within the discipline,⁴⁴ but the issues he addressed were real enough. In a series of papers he criticized psychologists' misuse and ignorance of the basis of statistical inference.⁴⁵ He wanted to make a distinction between "mathematical" and "scientific" significance, or between statistics and "scientific induction." By the latter he meant the exercise of the investigator's judgment about the psychological significance of a particular statistical finding, a judgment that would have to be made in the light of existing knowledge about the matter that was under investigation. What he deplored was the blind reliance on limited statistical samples, statistical conventions, and statistical assumptions for drawing substantial psychological conclusions: "Statistical ability divorced from a scientific intimacy with the fundamental observations leads nowhere."⁴⁶ The danger he sensed was that psychologists were beginning to confuse a constructed statistical reality with the psychological reality they were supposed to be investigating.

His concerns appear to have had little or no discernible influence on the investigative practices of the discipline. He was answered in no uncertain terms by T. L. Kelley, a key figure in the propagation of the idea that psychological research was essentially a statistical enterprise. Boring had summed up his argument as follows: "But, if in psychology we must deal – and it seems we must – with abilities, capacities, dispositions and tendencies, the nature of which we can not accurately define, then it is senseless to seek in the logical process of mathematical elaboration a psychologically significant precision that was not present in the psychological setting of the problem. Just as ignorance will not breed knowledge, so inaccuracy of definition will never yield precision of result."⁴⁷ On the contrary, maintained Kelley ("the error of believing that knowledge can be wrought out of ignorance is still strong within me"), that is exactly what a statistical psychology is able to achieve, and he proceeded to demonstrate what he meant by a number of examples taken from the field of mental measurement. His defense of arithmetic units as units of *mental* measurement is interesting: "Without denying the possibility of other workable systems, it seems that, in a civilization such as ours, steeped in the elementary associative and commutative principles of arithmetic and algebra, much is to be gained in simplicity and accuracy of interpretation if the units employed in mental measurement obey these well known laws."⁴⁸ What Kelley knew was what a later generation of psychologists was apt to forget –

namely, that the categories of quantitative psychology were intimately dependent on the social practices of a particular culture.

However, for Kelley this dependence was a positive feature, a source of strength. It made it easier to relate the results of psychological research to the practices of social institutions, as when measurement in terms of mental age units could be directly utilized by an age-graded system of elementary education. Beyond this, the adoption of statistical models made it possible to eliminate the element of "intuition" from psychological research and to replace it with strict logical inference. Boring, of course, had argued that this was both a dangerous and an impossible goal, but right from the start it was obvious that he represented a minority opinion. The use of statistical inference spread apace, and as it spread, its use became more rigidly circumscribed by conventions expressed in automatic procedures disseminated through the medium of statistical textbooks.⁴⁹

By the end of the 1930s certain practices in regard to the use of statistics in psychology had become well established. "Statistical significance" had become a widely accepted token of factual status for the products of psychological research. If a particular finding reached a conventionally fixed level of statistical significance, it was automatically received into the corpus of scientific psychological knowledge; if not, it was unlikely to be published, unless accompanied by other, "significant" findings.⁵⁰ At the same time, there was virtually no appreciation of the fact that commonly used statistical methods implied specific models of reality when applied to numerical data with psychological reference. These methods were looked upon as theory-free or, at a minimum, theory-neutral techniques whose employment was unproblematical, once a few conventions governing their use had been fixed. The commonly used textbooks of psychological statistics encouraged these attitudes by a cookbook approach to the subject, which avoided discussion of fundamental and controversial issues. In most cases the methods were presented anonymously, without human attribution, thus imparting to them a distinct aura of infallibility.⁵¹

These developments had certain undeniable advantages for the internal life of the discipline. In the first half-century of its existence, modern psychology had failed to generate any body of theory that was simultaneously precise, powerful (in the sense of wide applicability), and generally acceptable to most workers in the field. Yet, investigators continued to produce ambiguous data that were claimed to be scientific. That claim was difficult to justify on rational, theoretical grounds, given the lack of consensus on that level. The only alternative basis for that modicum of agreement that was necessary for saving the scientific status of the field was to be found on the level of methodology.⁵² Purely technical criteria could be used as a substitute for theoretical or substantive criteria in judging the value, and even the acceptability, of the products of psychological inves-

tigation. The ideal that began to emerge toward the end of our period was that of the automatization of the process of scientific inference. Although it might not be held together by substantial theoretical commitments a diverse group of investigators might still be able to function as a community of sorts if it adopted a specific set of unexamined methodological conventions.

One such convention that deserves special mention relates to the use of average data. As we saw in chapter 5, the averaging of quantitative information from many subjects was not part of the classical methodology of experimental psychology. However, with the demonstration of the usefulness of statistical data for many of the practical contexts in which psychology was beginning to be applied, such data gained a certain respect. When experimental methods were extended to the investigation of complex, molar behavior patterns – for which they had not originally been considered suitable – the use of average data seemed to provide an acceptable way of coping with the utter lack of consistency that was characteristic of complex individual behavior forced into the quantitative mold. Individual behavior might show little or no consistency from one occasion to another or from one experimental situation to another, but if data from many individuals were pooled, certain statistical regularities sometimes emerged. These regularities were to form the basis of psychological generalizations, even though the average pattern might not correspond to the actual behavior of a single individual member of the statistical group.⁵³

This trend was particularly noticeable in the new field of the psychology of “learning,” understood now in the very broad sense that men like Thorndike had given to the term and that had been adopted by functionalists like Carr as well as by behaviorists. The abstract psychological category of “learning” was intended to lead to the formulation of general laws of adaptive behavior that held for all animal organisms under all natural and artificial conditions. This was the field that, it was hoped, would provide a psychology by now strongly tied to the goal of behavior control, with the fundamental science on which its practical prescriptions would be based. A trickle of empirical studies of human and animal learning soon became a flood, the maze quickly establishing itself as the preferred physical setting for these investigations. However, it was not long before the gross inconsistency of individual performances in these experimental settings began to raise some doubts. A protracted discussion in the pages of the *Journal of Comparative Psychology* seems to have settled the issue for most of those working in this field.⁵⁴ Henceforth, the use of average or group data was to become the norm for this prestigious area of basic research. By 1935 an editorial statement in the same journal had to enter a gentle caveat against the pervasive use of “the average animal.”⁵⁵ Nevertheless, reliance on this fictional creature became even more pronounced. Finally,

it achieved the status of a methodological fetish, and the minority who refused to worship at its feet were forced to establish their own publication outlet, well out of the mainstream.⁵⁶

Two things are striking about this development. The first concerns the confusion between statistical issues and substantive psychological issues, which was present from the first, and the second concerns the ease with which statistical significance testing became accepted as an indispensable tool for the corroboration of psychological theories. These two features are not unrelated. What is noteworthy about the original confrontation of the issue of group data in the *Journal of Comparative Psychology* is that the issue is already defined as a statistical problem and only as a statistical problem. It is seen as a matter of handling intraindividual and interindividual *variability* on predetermined parameters. The question of the psychological appropriateness of these parameters and of the *psychological* significance of the measured variability never comes up. In particular, the requirement of the statistical model – that one deal with *continuous* magnitudes – was blindly accepted as binding on the underlying psychological reality. Thus, the possibility that the observed inconsistencies might be at least partly the result of underlying psychological discontinuities was not an alternative hypothesis that was available to these investigators.⁵⁷ For them there was no disjunction between the properties of the statistical aggregates they had constructed and the distribution of those psychological characteristics that had been drawn on to produce these artifacts. The problem of variability could therefore be reduced to one of how to test for *statistical* significance, conveniently ignoring the crucial question of what restrictions the statistical model implied by this practice imposed on the concepts of psychology.⁵⁸

Establishing the statistical significance of differences among aggregates was a practice of proven value in practical contexts where the overall effects of administrative decisions needed to be assessed. Given the prevailing tendency to reify statistical artifacts, and therefore to confuse statistical with psychological reality, it was quite natural for statistical significance testing to be employed as a basis for decisions about the validity of *psychological* hypotheses. Among the factors that helped to spread this practice, the automatization of scientific decision making, which it encouraged, has already been mentioned. More generally, it was a practice that reduced the demands made on psychological theorizing – no trivial achievement for a discipline that had never been able to get its theoretical house in order. Using statistical significance tests as the standard technique for the corroboration of psychological hypotheses meant that theories were generally regarded as “confirmed” if a very weak logical complement, “chance,” or “the null hypothesis” could be disconfirmed.⁵⁹ This travesty of scientific method certainly allowed the growth of a major research industry that offered employment to many, even though its products

can now be seen as having contributed nothing of either practical or theoretical value.⁶⁰

Not only did these methodological aberrations legitimate a large-scale waste of time, effort, and resources, they also confined psychological theorizing in an increasingly narrow mold, thus closing the door on alternative conceptualizations and practices that might have reversed the process of intellectual decline that the discipline was now beginning to suffer. Because methods were almost universally regarded as theory-neutral instruments, the assumptions on which they were based were able to function as unrecognized theoretical models for predefining basic features of the reality under investigation.⁶¹ The more rigidly the demands of a particular statistical methodology were enforced, the more effectively were ideas that did not fit the underlying model removed from serious consideration. Such ideas had first to be translated into a theoretical language that conformed to the reigning model before they could be seriously considered. In other words, they had to be eviscerated to the point where they no longer constituted a threat to the dominant system of preconceptions guiding investigative practices. The final stage of this process was reached when the statistical models on which psychologists had based their own practice were duplicated in their theories about human cognition in general.⁶²

One of the fundamental sociopsychological principles described by Wilhelm Wundt was based on what he called the "heterogony of ends."⁶³ By this he meant the tendency for human goals to have unintended consequences, which then modify the goals themselves. Such developments he tended to see as progress. The history of the discipline for whose paternity he was often held responsible certainly provided some striking illustrations of this principle. Fortunately for him he was spared the difficult task of reconciling the resulting developments with his rather touching nineteenth-century belief in the inevitability of progress.

10

Investigating persons

Extension of the mental test model

American psychologists, as we noted in chapter 7, had set about redefining the psychological problem of individual differences in terms of a comparison of performances against a competitively defined standard. This redefinition appeared to be perfectly adapted to the administrative requirements of social institutions whose efficient functioning depended upon some kind of selection process that was rationally defensible. The kinds of tasks that best lent themselves to the measurement of individual differences in this sense were tasks with answers that were unambiguously right or wrong. Performance on such tasks could be readily arithmetized by counting right and wrong answers, a procedure that had long been conventionalized in the institution of school examinations. Improvements in the precision of such procedures, made possible by Galton's fundamental contributions, eventually led to the construction of normed intelligence tests.

It is hardly a cause for surprise that educational institutions provided the setting in which intelligence tests proved themselves best adapted to administrative requirements. Transferred outside their home ground, their ability to predict future performance was much diminished, often to the point where their practical usefulness vanished altogether. This problem already emerged in the first major attempt at employing intelligence tests in the service of a noneducational institution, the U.S. Army in World War I. Most of the psychologists involved appear to have been convinced of the potential applicability of intelligence tests to military selection problems. However, although they were given considerable leeway and succeeded in testing well over a million men, the army does not seem to have found the results of this exercise particularly useful.¹ The military admin-